Sheet (3)

(1) Write a Python program that prompts the user to enter three integers and display these integers in ascending order.

(2) Suppose you shop for two different packages of rice. You would like to write a Python program to compare the cost. The program prompts the user to enter the weight and price of each package and displays the one with the better price.

(3) Write a Python program which randomly generates two, 3-digit integers, num1 and num2. asks the student a question such as "What is nun1+num2 ? " After the student enters the answer, the program displays a message indicating whether it is correct.

(4) Write a Python program which randomly generates two, 2-digit integers, num1 and num2. asks the student a question such as "What is nun1*num2 ? " After the student enters the answer, the program displays a message indicating whether it is correct.

(5) Write a program that prompts the user to enter the currency exchange rate between U.S. dollars (USD) and Egyptian pounds (EGP). Prompt the user to enter 0 to convert from U.S. dollars (USD) to Egyptian (EGP) and 1 for vice versa. Prompt the user to enter the amount in U.S. dollars or Egyptian EGP to convert it to Egyptian EGP or U.S. dollars, respectively. Here are some sample runs:

(6) You can use Cramer's rule to solve the following 2×2 system of linear equation:

$$a_1x + b_1y = c_1 \quad \text{and} \quad a_2x + b_2y = c_2,$$

$$x = \frac{c_1b_2 - b_1c_2}{a_1b_2 - b_1a_2} \qquad y = \frac{a_1c_2 - c_1a_2}{a_1b_2 - b_1a_2}$$

Write a Python program that prompts the user to enter a_1 , b_1 , c_1 , a_2 , b_2 and c_2 and displays the result. If $a_1b_2 - b_1a_2 = 0$, report that "The equation has no solution."

(7) The two roots of the quadratic equation $ax^2 + bx + c = 0$ can be obtained using the following formula: $R_1, R_2 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ where $b^2 - 4ac$ is called the *discriminant* of the quadratic equation. If it is positive, the equation has two distinct real roots. If it equals zero, the equation has two identical real roots. If it is negative, the equation has two complex roots.

Write a Python program that prompts the user to input the coefficients a, b, and c, evaluates the discriminant, determines the nature of the roots, and then computes and prints the roots accordingly.

(8) Write a Python program that reads the three sides of a triangle (a , b , and c) and computes the area if the input is valid. Otherwise, it displays that the input is invalid. The formula for computing the area of a triangle is

$$area = \sqrt{s(s-a)(s-b)(s-b)}$$

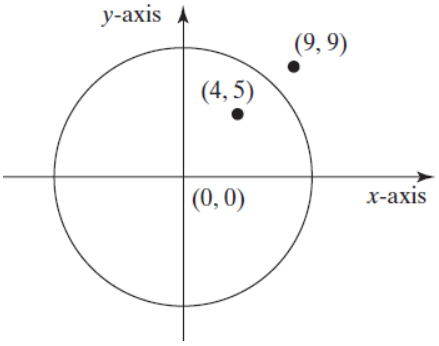
Where $s = (a+b+c)/2$

(9) Given the lengths of three sides of a triangle (a , b , and c), its three interior angles can be calculated using the Law of Cosines:

$$A = \cos^{-1}\left(\frac{a^2-b^2-c^2}{-2bc}\right),\ B = \cos^{-1}\left(\frac{b^2-a^2-c^2}{-2ac}\right),\ C = \cos^{-1}\left(\frac{c^2-b^2-a^2}{-2ab}\right)$$

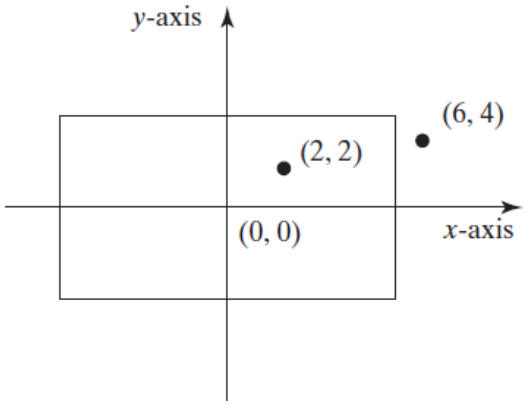
Write a Python program that accepts three side lengths as user input. The program must first verify whether the given sides can form a valid triangle using the triangle inequality theorem. If the triangle is valid, calculate and print its three interior angles in degrees.

(10) Write a Python program that prompts the user to enter a point $(x.y)$ and checks whether the point is within the circle centered at (0.0) with radius 10. For example, (4.5) is inside the circle and (9.9) is outside the circle, as shown in in the below figure.



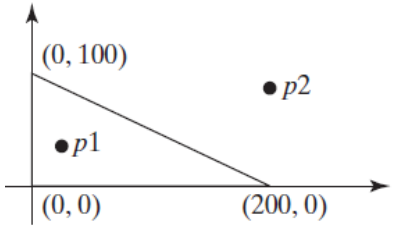
(**Hint:** A point is in the circle if its distance to (0.0) is less than or equal to 10. The formula for computing the distance is $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$. Test your program to cover all cases.)

(11) Write a Python program that prompts the user to enter a point $(x.y)$ and checks whether the point is within the rectangle centered at (0.0) with width 10 and height 5. For example, (2.2) is inside the rectangle and (6.4) is outside the rectangle, as shown in the below figure.

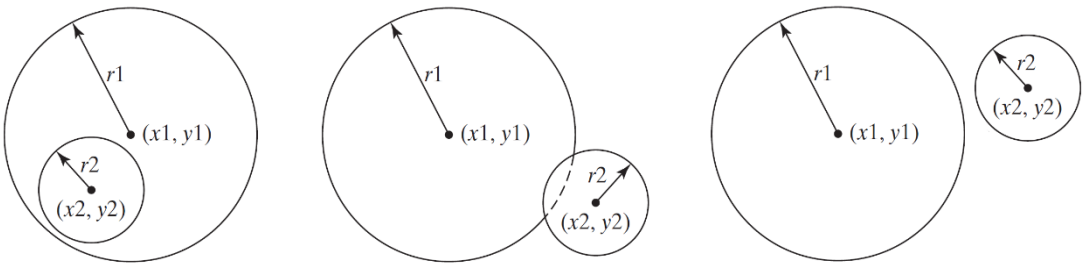


(**Hint:** A point is in the rectangle if its horizontal distance to (0.0) is less than or equal to $10/2$ and its vertical distance to (0.0) is less than or equal to $5/2$. Test your program to cover all cases.)

(12) Suppose a right triangle is placed in a plane as shown below. The right-angle point is placed at (0.0), and the other two points are placed at (200.0), and (0.100). Write a Python program that prompts the user to enter a point with x- and y-coordinates and determines whether the point is inside the triangle.



(13) Write a Python program that prompts the user to enter the center coordinates and radii of two circles and determines whether the second circle is inside the first or overlaps with the first, as shown in the below figure.



(**Hint:** circle2 is inside circle1 if the distance between the two centers $\leq |r1 - r2|$ and circle2 overlaps circle1 if the distance between the two centers $\leq |r1 + r2|$. Test your program to cover all cases.)

(14) Body Mass Index (BMI) is a screening tool that estimates body fat based on a person's weight and height. It is calculated by dividing weight (in kilograms) by height squared (in meters). That is

$$BMI = \frac{Weight\ in\ Kg}{(Height\ in\ m)^2}$$

Write a Python program that prompts the user to enter a person's weight in kilograms and height in meters. The program calculates the BMI and gives an interpretation for the given data according the following table;

BMI	Interpretation
$BMI < 18.5$	Underweight
$18.5.0 \leq BMI < 25.0$	Normal
$25.0 \leq BMI < 30.0$	Overweight
$30.0 \leq BMI$	Obese